

How the app works

Every mapping app will tell a World Cup visitor the *fastest* way from one place to another. Not one of them will tell them which way has shade. Here's what we built instead — what it measures, how it picks a winner, and every number behind it.

01

The problem it solves

Houston hosts seven FIFA World Cup 26 matches at NRG Stadium in June and July. On a typical afternoon in those months the heat index — what the temperature actually feels like once humidity is accounted for — regularly clears 105 °F. In those conditions, the fastest way somewhere is frequently not the way anybody should take.

A normal router optimises for time, so it'll happily send someone down a treeless feeder road beside six lanes of traffic to save ninety seconds. You can guess what happens next: the visitor takes one look at that walk, decides it's unbearable, and calls a car — which is precisely the outcome a sustainability project exists to prevent.

So we asked a different question. Given two points in Houston, the app builds several realistic ways to get between them, then judges each one on whether a person would actually be willing to make that trip.

02

What you get

You type a start and a finish, choose how you are travelling, and press one button.

Back comes a set of genuinely different routes — up to six — each with a score out of 100, the time and distance to expect, and the things that trip will actually cost you. They're all drawn on the map at once, so you can compare them at a glance. Pick one and the detail fills in: turn-by-turn directions, where the shade is, where the drinking fountains are, and what the weather will be doing while you're out in it.

0–100

A pleasantness score for each route, built from five measurements whose balance you control

4

Ways to travel: on foot, by bike, by car, or on METRORail and METRO buses

6

Live data sources behind every answer — none of it typed into the app by hand

Alongside the score sit the sustainability and comfort figures: the carbon emitted, or avoided compared with driving the same trip alone; the calories burned; the cost; and two numbers specific to a Houston summer — how many **minutes you spend outdoors with no shade over your head**, and the **longest stretch of the trip with no drinking water**.

What the score is made of

Five inputs go into the number on each route card. One of them dominates, and one of them deliberately counts for nothing — and that second decision is the more interesting one.

■ Walkability — 55%

A ground-level index of how hard a place is to walk, built by our GIS team from 30-metre satellite and LiDAR data covering all of Harris County. It folds in tree canopy, urban heat, road context and more. It's far and away the biggest thing in the score, because it's far and away the best data we have.

■ Directness — 25%

Does the route actually get you there, or does it wander? This takes the biggest remaining share for a simple reason: it's the one thing the walkability index can't see. The index describes a *place*. Directness is a property of your *route* across it.

■ Green space — 12%

How much of the walk runs inside or within 60 metres of a park, bayou or garden. This is about pleasantness rather than shelter — being beside a bayou trail is nicer than being between two parking lots, whether or not there's a tree over you.

■ Away from traffic — 8%

How much of the distance runs along freeways, tollways and feeder roads. It's kept small on purpose: the walkability index already knows about road context, so weighting this heavily would count the same six lanes twice.

■ Tree canopy — 0%

Reported everywhere, scored nowhere. This is the decision worth explaining, and it's explained below.

WHY SHADE COUNTS FOR ZERO IN AN APP ABOUT SHADE

It looks like a contradiction. It isn't. The GIS index already contains tree canopy and urban heat, measured from LiDAR and satellite imagery at 30-metre resolution across the whole county. Our own shade layer is built from about two thousand individually mapped trees in OpenStreetMap.

So these aren't two views of the same thing at different resolutions. One is simply better. Scoring both would count canopy twice — and the second time, badly. So the good measurement does the scoring, and ours does the reporting: it's still what drives "unshaded minutes" and the shade row on the route profile, because it's the only canopy signal we can pull out on its own. If the GIS team ever exports their canopy layer separately, theirs replaces ours outright.

You can move all five. The sliders are right there on the map, and the ranking re-sorts as you drag — including the canopy slider, which starts at zero and is labelled with exactly what raising it does: double-count the trees.

Telling it where you are going

There are five ways to set either end of a trip. You can type and pick from live suggestions; click the map after pressing **Set start** or **Set finish**; drag either pin to move it; paste a pair of coordinates; or press the locate button to use where you actually are. Clicking or dragging looks the place up in reverse, so the box shows a real name rather than a pair of numbers.

The search covers **all of Texas but favours Houston**, so nearby results rise to the top without out-of-town places being excluded. The app also understands about **forty local abbreviations** — the shorthand Houstonians actually type, which no geocoding service recognises on its own — and expands them before searching.

Two limits apply before anything is planned. The two points must be at least **25 metres** apart, and the trip must be within reach of the mode: **42 km** walking, **120 km** cycling or by METRO, **800 km** driving.

How it finds routes

This is the part that makes the app more than a map with a line drawn on it.

Here's the thing about standard routing services: they only ever offer the two or three *fastest* options. In a grid city like Houston those come back near-identical — the same roads in a slightly different order — which gives you nothing worth comparing. So the app goes and builds its own candidates, fresh for every trip:

1 Ask the routing service for the fast options.

Up to three time-optimal routes, then de-duplicated, since routers often return the same road twice with one junction changed.

2 Download what is around the trip.

Every park, wood, bayou, tree and drinking fountain in the corridor between the two points — a box drawn around the routes and padded by 2 km.

3 Find parks worth a detour.

Green spaces of at least **2.5 hectares** that would add no more than the greater of **1.2 km** or **45%** of the straight-line distance. The best three are taken, kept at least 700 m apart so three routes don't run through the same park.

4 Route the trip through each of them.

These are real, routable trips added to the list, not sketches.

5 Trim the field, then measure what is left.

Anything more than **1.9×** the shortest option is dropped, and at most six routes are scored and shown.

WORTH SAYING PLAINLY

The route this app calls "most pleasant" is typically one no routing engine would ever have offered you. That is the point of the project, not a side effect of it.

How a route gets measured

The app walks each route in small steps — a check **every 75 metres** — and at every one of those points it asks the same handful of questions. *How hard is this ground to walk? Am I in or beside green space? Are there trees over me? Am I on a big road?*

A two-mile route picks up a couple of hundred of these checks, and the percentages you see are simply how often the answer came back yes. Because every individual answer is kept rather than just the running total, the app can also show you *where* along the route each condition held. That's the profile chart — and it's why the chart can never disagree with the percentage printed above it. They're the same data, averaged and unaveraged.

The score, and two ways to read it

score = 100 × (0.55 × walkability + 0.25 × directness + 0.12 × green + 0.08 × low traffic + 0 × canopy)

Each input is a value between 0 and 1. Those are the starting weights; the sliders change them, and whatever you set is rebalanced to total 100%.

One detail that took a while to get right. Walkability is scored as ease *per unit of progress toward your destination*, not as an average along the way — and the difference matters more than it sounds. Averaging quality over a route is blind to length, so a detour through a park raises the average while making you walk 70% further. Early on, that meant the longest route kept winning. Multiplying by how directly the route gets you there fixes it: a route now has to be pleasant *and* actually take you somewhere.

Where the walkability surface has no coverage, that input drops out and the remaining weights are rebalanced, so a route is never punished for leaving the study area. The score stays comparable — it just rests on less evidence, and the app says so.

The same route can then be scored on two scales, and a toggle switches between them instantly.

	ABSOLUTE	RELATIVE
What it means	The real share of the route that's green, walkable and off big roads. A 60 today is the same 60 tomorrow, on any trip.	Each measurement stretched across the options currently on screen, so the best scores 1 and the worst scores 0.
Good for	Comparing different trips, tracking change over time, reporting a number that stands alone.	Choosing between options that look nearly identical.
Behaves like	A measurement. Nothing quite reaches 100.	A ranking. Something always reaches 100.

They can pick different winners on the same set of routes, which is exactly why both are there rather than one being chosen for you. Five routes also get badged automatically: **Fastest**, **Shortest**, **Most pleasant**, **Greenest** and **Most shaded**, plus a water badge when there are fountains to find.

The heat you are actually in

Every trip is planned at a specific time, and the conditions for those hours come from an hourly forecast covering the next **seven days**. That turns an abstraction into something concrete:

Feels like 106 °F (Danger) — air 94 °F at 55% humidity. You are outside and unshaded for 44 minutes of it, under UV 7.

The "feels like" number is calculated here, not taken from the forecast

The weather service hands us its own apparent temperature, and we deliberately ignore it. If a number is going to carry this much weight, we ought to be able to show our working — so the app computes the **US National Weather Service heat index** itself, from measured temperature and humidity, using the NWS's published equation. It lands within about a degree of the forecast service's own figure, and it's the number Houstonians hear on the news.

The result is placed in one of the five official warning bands, which is what the colour on screen encodes — the bands are what heat warnings actually get issued against, and each is named in words beside its colour.

HEAT INDEX	BAND	WHAT IT MEANS
Below 80 °F	No heat caution	Nothing to plan around
80 – 90 °F	Caution	Fatigue possible with prolonged exposure
90 – 103 °F	Extreme Caution	Heat cramps and heat exhaustion possible
103 – 125 °F	Danger	Heat exhaustion likely, heat stroke possible
Above 125 °F	Extreme Danger	Heat stroke highly likely

Choosing a better hour

Below the summary is a strip of the coming hours, coloured by band. It is also a control: click an hour and the trip is re-planned for it.

Leaving at 8pm is 11 °F easier. This trip would feel like 95 °F then, against 106 °F now. Same route, same shade — different afternoon.

Two rules keep that suggestion useful. It appears only when the heat index is **90 °F or above** and the better hour is **within six hours**; unbounded, the advice degenerates into "travel at night", which is true of every hot place and helps nobody. And both temperatures are averaged over the hours the trip actually occupies rather than read off the departure time, so a ninety-minute journey is not compared against a single hour it is only partly in.

Drivers get the conditions but no suggestion, because a driver is not in the weather. Air quality rides along too: Houston's signature pollutant peaks on hot, still afternoons, and it matters most to whoever is breathing hardest — so it is raised only when the air quality index passes **100**, the EPA's "unhealthy for sensitive groups" line, and you are walking or cycling rather than driving.

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Taking METRO

Choose METRO and the app plans a genuine trip on Houston's network — METRORail and the buses, with the route numbers, stop names, headsigns and departure times a rider will actually see. **None of that is written into the app.** There is no timetable in the source to fall out of date: every number and time is read live from METRO's own published schedule data, so if METRO reroutes a bus tomorrow, the app follows without anyone touching it.

Three searches, not one

Asked for the best way from one place to another, a transit planner returns the optimal set for the departure window — which in Houston is usually the same rail line repeated at every headway. True, and useless for comparing anything. So the app asks three times: **anything, buses only, and rail only**, each over a **one-hour** departure window returning up to four itineraries. Together they surface the alternatives the optimal answer hides.

Results are then folded together by which routes they use, so you get one entry per genuinely different trip rather than eleven copies of the same train. Asked to leave at a time, the app keeps the earliest departure of each distinct trip; asked to arrive by a time, the latest one that still makes it. Anything more than **three hours** from the time you asked about is discarded.

A transit trip is scored on the part you are outside for

This is where our own thinking had to change. Average shade over an entire transit trip and what you've measured is the tree cover along a rail corridor, seen through a window — a number about nobody's comfort. Worse, a long ride drowns out the short walk that actually decides whether the trip is bearable. So for a METRO trip:

Green and shade	Measured over the walking legs only. The cards say "Green on foot".
Unshaded minutes	Counts the walk to the stop <i>plus the wait at it</i> , and not the ride. A 52-minute rail trip with a 9-minute walk and a 6-minute wait leaves you in the sun for about fifteen minutes, and that is the number reported.
Distance from traffic	Measured on the walking. Whether the bus runs down a feeder is not a question that applies from inside it.
Drinking water	Matched only against the parts you walk, with the dry stretch measured within each, so the ride between two walks is never counted as a gap between fountains.
Carbon	Charged leg by leg — the walking emits nothing, and a train and a bus are counted at their own rates.
Route profile	Gains a fourth row showing where you are on board, so the trip does not read as miles of unshaded pavement.

Step-free routing

Tick **step-free routes only** and the walk to the stop, every transfer and the walk off are all planned at wheelchair pace — about **0.69 metres per second** against **1.03** on foot — rather than filtering the vehicles, since METRO's fleet is marked accessible throughout and METRORail has level boarding at every platform. On most Houston trips the step-free route is therefore the same route, simply walked more slowly, and the extra time is time spent outdoors.

Each pace is given a walking budget that buys it the same distance — **20 minutes** on foot, **30 minutes** at wheelchair pace, both about 1.2 km — so the two are compared over the same ground. A second, unrestricted search runs alongside purely to price the difference, letting the app state what step-free costs on this particular trip in minutes and transfers.

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Water

Drinking fountains **within 120 metres** of the route are matched to it, pinned on the map, and listed in the order you would reach them, each with how far along the route it sits. Fountains mapped within 40 metres of one another are treated as one, since parks often map several a few paces apart, and decorative fountains are excluded because they are not drinkable.

The headline number is not how many exist but **the longest stretch without one** — the question that actually matters at 100 °F. It includes the gaps at either end, so a route with a fountain in the middle and nothing for the first mile reports that first mile.

Seeing where, not just how much

A percentage tells you how much of a route is shaded. It can't tell you *which part* — and when you're deciding whether to make the walk, that's usually what you want to know. Two features answer it.

The **route profile** is like an elevation chart, except the rows show conditions instead of hills: where the route is green, where it is shaded, where it runs beside traffic, and — on a transit trip — where you are on board. Water stops are marked above it and turns ticked below, on the same scale. Hovering reads out the conditions at that point, and clicking flies the map there, which turns the chart into a scrubber for the route.

The **directions** are ordinary turn-by-turn instructions with one addition: each step is re-measured at **20-metre** resolution, so the list tells you which particular turns are shaded, which are green, and which put you on a feeder road. Clicking an instruction zooms the map to that manoeuvre. Steps under **25 metres**, and "continue" steps that merely restate the road you are already on, are folded into the previous instruction — which turns a noisy 75-step walking route into about 50 usable ones.

Two behaviours that are deliberate

Nothing searches until you press the button

A comparison is expensive: it involves several requests to free, shared, public services. Earlier versions searched automatically whenever anything changed, which meant setting up a single trip could fire off four searches before you had finished describing it.

Now only the button searches. Everything else reports what changed and leaves the existing results on screen — flipping to another mode to look and flipping back should not cost a fresh search — but those results carry an "outdated" marker so a stale list is never mistaken for a current one. Anything answerable from information already in hand still happens instantly: moving the weight sliders, switching scoring scales, or picking a different hour for a walking, cycling or driving trip. Only METRO returns to the network for a new hour, because its timetable genuinely differs by the hour.

On a phone

Below **900 pixels** wide, the side panel becomes a sheet you drag up from the bottom over a full-screen map, with three sizes: a peek, half, and full. Minimised it shows just the trip and the headline result. When you select a route the map frames it against the part of the screen you can actually see rather than the middle of the map — which is the difference between seeing your route and seeing the sheet sitting on top of it.

Every number the app uses

Each figure below is either a setting the app measures with, or a published average applied to the distance travelled. They are listed on screen as well, so provenance travels with the number.

MEASURING A ROUTE

Checks taken along a route	every 75 m	Roughly 200 points on a two-mile trip
Counts as beside green space	within 60 m	Of a park, bayou, garden, wood or cemetery
Counts as under canopy	within 30 m	Of a mapped tree or line of trees
Counts as a water stop	within 120 m	Of the route; duplicates merged within 40 m
Directions re-measured at	every 20 m	So each individual turn gets its own reading
Turn-heavy penalty reaches zero at	15 turns / km	Applied to the directness component

SCORING

Walkability weight	55%	The GIS index — the largest single input
Directness weight	25%	The one thing the index cannot see
Green space weight	12%	Park and bayou frontage
Away-from-traffic weight	8%	Kept small; the index knows about roads already
Tree canopy weight	0%	Reported, not scored — the index measures it better
All five	adjustable	Sliders on the map; always rebalanced to 100%

BUILDING THE ROUTE OPTIONS

Fast options requested	3	From the routing service, then de-duplicated
Smallest park worth a detour	2.5 hectares	Below this, rerouting is not worth it
Largest detour accepted	1.2 km or 45%	Whichever is greater, over the straight-line distance
Park detours generated	up to 3	Kept at least 700 m apart from one another
Longest route kept	1.9× shortest	Anything longer is discarded before scoring
Routes scored and shown	up to 6	The final comparison set
Map data area	route + 2 km	Padded to 400 m for the walking parts of a METRO trip

TRIP LIMITS

Minimum trip length	25 m	Below this, start and finish are the same place
Maximum walking trip	42 km	Roughly a marathon
Maximum cycling trip	120 km	A long day's ride
Maximum METRO trip	120 km	About as far as the network reaches
Maximum driving trip	800 km	Beyond this, nothing is routed

METRO

Searches run per trip	3	Anything, buses only, rail only
Departure window searched	1 hour	Long enough to catch a second option on a 30-minute headway
Itineraries per search	up to 4	Before de-duplication by routes used
Results kept within	3 hours	Of the time you asked about
Walking pace	1.03 m/s	Standard pedestrian profile
Wheelchair pace	0.69 m/s	Step-free profile
Walking budget, on foot	20 min	About 1.2 km to or from a stop
Walking budget, step-free	30 min	The same 1.2 km at wheelchair pace

HEAT AND AIR

Forecast horizon	7 days	Hourly temperature, humidity, UV and rain
Air quality horizon	5 days	Hourly ozone, fine particulates and index
Cooler-hour advice appears at	90 °F	Heat index, and only within the next six hours
Air quality raised at	index 100	The EPA's "unhealthy for sensitive groups" line
Shade is worth about	15 °F	The gap between full sun and the shaded heat index

IMPACT FIGURES

Driving carbon	251 g / km	US EPA typical passenger vehicle, ~404 g per mile, one occupant
Bus carbon	105 g / km	Federal Transit Administration averages, local bus at average occupancy
Rail carbon	59 g / km	Scaled from the bus figure by the FTA's light-rail to bus ratio
Cycling carbon	5 g / km	European Cyclists' Federation, manufacturing and maintenance share
Walking carbon	0 g / km	No vehicle; the energy is reported as calories instead
Calories, walking	62 kcal / km	About 100 kcal per mile for a 70 kg adult
Calories, cycling	30 kcal / km	About 500 kcal per hour at a 16 km/h pace
Driving cost	\$0.42 / km	IRS standard mileage rate — fuel, maintenance, insurance, depreciation
Cycling cost	\$0.03 / km	Maintenance allowance

FRESHNESS AND PATIENCE

Map data reused for	24 hours	Parks, trees and fountains, cached per area
Transit results reused for	5 minutes	Identical trips are not re-asked
Forecast reused for	30 minutes	Cached per area of the city
Place search gives up after	12 seconds	Then falls back to the next option
Route request gives up after	20 seconds	Per attempt
Map data and transit give up after	25 seconds	Then the next mirror is tried
Basemap switches after	10 failures	Automatically, to the next provider

Where the data comes from

Routes for walking, cycling, driving	OSRM, an open-source routing engine, on free public infrastructure
METRO routes, stops and times	Houston METRO's own published schedule feed, routed live by MOTIS via Transitous
Parks, trees, fountains	OpenStreetMap, queried through Overpass
Temperature, humidity, UV, air quality	Open-Meteo, drawn from national weather models
Heat index	Calculated in the app from temperature and humidity, using the US National Weather Service equation
Carbon, calories, cost	US EPA, Federal Transit Administration, European Cyclists' Federation and IRS published figures, applied to the routed distance
Green %, shade %, unshaded minutes, water gaps	Calculated in the app from map geometry along each route

Built for the FIFA Sustainability Hackathon. Routing by OSRM; METRO trips by MOTIS via Transitous over Houston METRO's published GTFS feed; map data © OpenStreetMap contributors via Overpass; weather and air quality from Open-Meteo, with the heat index computed in-app using the US National Weather Service equation.